

Exercise 2

Arham Lodha

March 05, 2025

Problem 1a.

Proof. $\forall A \in SL_2(\mathbb{Z})$ and for all μ -integrable functions $f : \mathcal{H} \rightarrow \mathbb{C}$. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let $\mathbb{R}_+^2 := \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$.

$$\begin{aligned}
 u + iv = w = A \cdot z &= \frac{az + b}{cz + d} \\
 &= \frac{a(x + iy) + b}{c(x + iy) + d} \\
 &= \frac{((ax + b) + iay)((cx + d) - icy)}{|cz + d|^2} \\
 &= \frac{((ax + b)(cx + d) + acy^2) + iy(-acxy - bc + ad + acxy)}{|cz + d|^2} \\
 &= \frac{((ax + b)(cx + d) + acy^2) + iy}{|cz + d|^2}
 \end{aligned}$$

Note $z \mapsto A \cdot z$ is holomorphic on $\mathcal{H}$. The determinant of the jacobian (since for holomorphic g determinant of jacobian is $|g'|^2$) is

$$\begin{aligned}
 \left| \frac{d}{dz} \left(\frac{az + b}{cz + d} \right) \right|^2 &= \left| \frac{a(cz + d) - (az + b)c}{(cz + d)^2} \right|^2 \\
 &= \left| \frac{1}{(cz + d)^2} \right|^2 \\
 &= \frac{1}{(cz + d)^2 (cz + d)^2} \\
 &= \frac{1}{|cz + d|^4}
 \end{aligned}$$

Thus

$$dudv = \frac{1}{|cz + d|^4} dx dy$$

. Thus

$$d\mu(w) = \frac{dudv}{v^2} = \frac{dxdy}{|cz+d|^4 \frac{y^2}{|cx+d|^4}} = \frac{dxdy}{y^2} = d\mu(z) = d\mu(A^{-1} \cdot w)$$

$$\begin{aligned} \int_{\mathcal{H}} f(A \cdot z) d\mu(z) &= \int_{\mathcal{H}} f(w) d\mu(A^{-1} \cdot w) \\ &= \int_{\mathcal{H}} f(w) d\mu(w) \end{aligned}$$

Thus μ is invariant under the $SL_2(\mathbb{Z})$ action. $\square$

Problem 1b.

Proof. Let $f : \mathcal{H} \rightarrow \mathbb{C}$ be modular of weight of $k \in \mathbb{Z}$. Let $\varphi : \mathcal{H} \rightarrow \mathbb{C}$ where $z \rightarrow |f(z)| \operatorname{Im}(z)^{\frac{k}{2}}$. $\forall A \in SL_2(\mathbb{Z})$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\forall z \in \mathcal{H}$:

$$\begin{aligned} \varphi(A \cdot z) &= |f(A \cdot z)| \operatorname{Im}(A \cdot z)^{\frac{k}{2}} \\ &= |(cz+d)^k f(z)| \left(\frac{\operatorname{Im}(z)}{|cz+d|^2} \right)^{\frac{k}{2}} \\ &= \frac{|cz+d|^k}{|cz+d|^k} |f(z)| \operatorname{Im}(z)^{\frac{k}{2}} \\ &= |f(z)| \operatorname{Im}(z)^{\frac{k}{2}} \implies \varphi \text{ is modular of weight } 0 \end{aligned}$$

$\square$

Problem 1c.

Proof. Suppose f is meromorphic on $\mathcal{H}$ and modular of weight $k \geq 2$.

$\implies$: Suppose f is a cusp form. Thus $f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$ and $a_0 = 0$. $\forall x \in \mathbb{R}$

$$\lim_{y \rightarrow \infty} f(x+iy) = \lim_{y \rightarrow \infty} \left(\sum_{n=1}^{\infty} a_n e^{2\pi i n x} e^{-2\pi n y} \right) = 0$$

Thus as $y \rightarrow \infty$, $f(x+iy) \rightarrow 0$ and $|f(x+iy)| \rightarrow 0$ exponentially. Moreover for large y , $|f(x+iy)| \leq C e^{-2\pi y}$. Thus

$$\begin{aligned} \varphi(x+iy) &= |f(x+iy)| y^{\frac{k}{2}} \\ &= C e^{-2\pi i y} y^{\frac{k}{2}} \end{aligned}$$

Exponential decay dominates polynomial growth so as $y \rightarrow \infty$, $\varphi(x+iy) \rightarrow 0$ $\square$

Problem 2a.

Proof. Let $f : \mathcal{H} \rightarrow \mathbb{C}$ where

$$f(z) = \prod_{n \geq 1} (1 - e(nz))$$

. Let $g : \mathcal{H} \rightarrow \mathbb{C}$ where

$$g(z) = \prod_{n \geq 1} (1 + |-e(nz)|) = \prod_{n \geq 1} (1 + |e^{-2\pi n \text{Im}(z)}|) = \prod_{n \geq 1} (1 + e^{-2\pi n \text{Im}(z)})$$

. Redefine $g : \mathcal{H} \rightarrow \mathbb{C}$ to $g : (0, \infty) \rightarrow \mathbb{R}$ where

$$g(y) = \prod_{n \geq 1} (1 + e^{-2\pi ny})$$

. Note f is absolutely uniformly convergent iff g is.

Note, $\forall n \in \mathbb{N}$ and $\forall y \in (0, \infty)$, $0 < e^{-2\pi ny} < 1 \implies e^{-2\pi ny} \neq -1$. Additionally note since $e^{-2\pi ny} > 0 \implies \ln(1 + e^{-2\pi ny}) < e^{-2\pi ny}$

$$\begin{aligned} \ln(g(y)) &= \sum_{n \geq 1} \ln(1 + e^{-2\pi ny}) \\ &\leq \sum_{n \geq 1} e^{-2\pi ny} \end{aligned}$$

$\sum_{n \geq 1} e^{-2\pi ny}$ is uniformly convergent thus absolutely uniformly convergent. Thus $\ln(g(y))$ is absolutely uniformly convergent by Wierstrass M Test thus $g(y)$ is absolutely uniformly convergent thus $f(z)$ is absolutely uniformly convergent. Thus $(f(z))^{24}$ is absolutely uniformly convergent and thus $\Delta(z)$ is absolutely uniformly convergent since $e(z)$ is entire and trivially absolutely uniformly convergent.

Furthermore $e(z)$ and $(1 - e(nz))$ are holomorphic on $\mathcal{H}$, thus $\Delta(z)$ is holomorphic on $\mathcal{H}$ $\square$

Problem 2b.

Proof. Note the zeros of $\sin(z)$ are $k\pi$ for $k \in \mathbb{Z}$. Since $\cot(z)$ is the ratio of two entire functions $\cos(z)$ and $\sin(z)$, it is holomorphic on $\mathbb{C} - \{\text{zeros of } \sin(z)\} = \mathbb{C} - \{k\pi : k \in \mathbb{Z}\}$. Furthermore, $\frac{1}{\cot(k\pi)} = \tan(k\pi) = \frac{0}{(-1)^k} = 0$. Thus $\{k\pi : k \in \mathbb{Z}\}$ are the only points in $\mathbb{C}$ where $\cot(z)$ isn't holomorphic and its poles. Thus $\cot(z)$ is meromorphic on $\mathbb{C}$.

Note

$$\begin{aligned} \cot(z) &= \frac{\cos(z)}{\sin(z)} \\ &= \frac{\cos(z)}{\sum_{n=0}^{\infty} \frac{(-1)^{n+k}(x-k\pi)^{2n+1}}{(2n+1)!}} \\ &= \frac{\cos(z)}{(x-k\pi) \sum_{n=0}^{\infty} \frac{(-1)^{n+k}(x-k\pi)^{2n}}{(2n+1)!}} \end{aligned}$$

Note $\sum_{n=0}^{\infty} \frac{(-1)^{n+k}(x-k\pi)^{2n}}{(2n+1)!}$ is not 0 around a neighborhood of $k\pi$. So $\frac{\cos(z)}{\sum_{n=0}^{\infty} \frac{(-1)^{n+k}(x-k\pi)^{2n}}{(2n+1)!}}$ is holomorphic in a neighborhood around $k\pi$. Thus $k\pi$ is a pole of order 1.

Thus

$$\begin{aligned}
\operatorname{res}_{k\pi}(\cot) &= \lim_{z \rightarrow k\pi} (z - k\pi) \frac{\cos(z)}{\sin(z)} \\
&= \lim_{z \rightarrow k\pi} \frac{\cos(z)}{\sum_{n=0}^{\infty} \frac{(-1)^{n+k} (z - k\pi)^{2n}}{(2n+1)!}} \\
&= \frac{\cos(k\pi)}{(-1)^k} = \frac{(-1)^k}{(-1)^k} = 1
\end{aligned}$$

$$\begin{aligned}
\cot(z) &= i \left(\frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}} \right) \\
&= i \left(\frac{e^{iz}(1 + e^{-2iz})}{e^{iz}(1 - e^{-2iz})} \right) \\
&= i \left(\frac{1 + e^{-2iz}}{1 - e^{-2iz}} \right) \\
&= -i \left(\frac{-1 - e^{-2iz}}{1 - e^{-2iz}} \right) \\
&= -i \left(\frac{1 - e^{-2iz} - 1 - 1}{1 - e^{-2iz}} \right) \\
&= -i \left(1 - \frac{2}{1 - e^{-2iz}} \right)
\end{aligned}$$

□

Problem 2c.

Proof. Note $\tau \in \mathcal{H}$ fixed.

$$\begin{aligned}
f_m(z) &:= \cot \left(\left(m + \frac{1}{2} \right) z \right) \cot \left(\left(m + \frac{1}{2} \right) z / \tau \right) \\
g_m(z) &:= \frac{f_m(z)}{z}
\end{aligned}$$

1. For $k \in \mathbb{Z} \setminus \{0\}$:

$$\begin{aligned}
\frac{1}{g_m \left(\frac{\pi k}{m + \frac{1}{2}} \right)} &= \frac{\pi k \sin(\pi k) \sin(\frac{\pi k}{\tau})}{(m + \frac{1}{2}) \cos(\pi k) \cos(\frac{\pi k}{\tau})} \\
&= 0
\end{aligned}$$

Thus $\frac{k\pi}{m+0.5}$ is a pole of $g_m(z)$.

Let $a - bi = \tau^{-1}$.

$$\begin{aligned}\sin\left(\frac{k\pi}{\tau}\right) &= \sin(k\pi a + ik\pi b) \\ &= \sin(k\pi a) \cosh(k\pi b) - i \cos(k\pi a) \sinh(k\pi b)\end{aligned}$$

This is only 0, if $\sin(k\pi a) \cosh(k\pi b) = 0$ and $\cos(k\pi a) \sinh(k\pi b) = 0$. Since $\cosh(x) \geq 1$ and $\sinh(x) > 0$. Then $\sin(k\pi a) = 0 = \cos(k\pi b)$. This is a contradiction. Thus $\sin\left(\frac{k\pi}{\tau}\right) \neq 0$. Thus $\cot\left((m + \frac{1}{2})z/\tau\right)$ is homomorphic in a small neighborhood around $\frac{k\pi}{m+0.5}$. Furthermore $\cot\left((m + \frac{1}{2})z/\tau\right)/z$ is homomorphic around $\frac{k\pi}{m+0.5}$.

$$\frac{d}{dz} \left(\tan\left((m + \frac{1}{2})z\right) \left(\frac{k\pi}{m + \frac{1}{2}}\right) \right) = (m + \frac{1}{2}) \frac{1}{\cos^2(k\pi)} = (m + \frac{1}{2}) \neq 0$$

Thus $k\pi/m + \frac{1}{2}$ is a 1st order pole of $\cot((m + \frac{1}{2})z)$ and thus is a 1st order pole of $g_m(z)$

2. For $k \in \mathbb{Z} \setminus \{0\}$:

$$\begin{aligned}\frac{1}{g_m\left(\frac{\pi k\tau}{m+\frac{1}{2}}\right)} &= \frac{\pi k \sin(\pi k\tau) \sin(\pi k)}{(m + \frac{1}{2}) \cos(\pi k\tau) \cos(\pi k)} \\ &= 0\end{aligned}$$

Thus $\frac{\pi k\tau}{m+\frac{1}{2}}$ are poles of g_m .

Note $\sin(k\pi\tau) \neq 0$ for a similar argument to above. Thus $\cot((m + \frac{1}{2})z)/z$ is holomorphic in a neighborhood around $\frac{k\pi\tau}{m+0.5}$.

$$\frac{d}{dz} \left(\tan\left((m + \frac{1}{2})z/\tau\right) \left(\frac{k\pi\tau}{m + \frac{1}{2}}\right) \right) = (m + \frac{1}{2})/\tau \frac{1}{\cos^2(k\pi)} = (m + \frac{1}{2})/\tau \neq 0$$

Thus $k\pi\tau/m + \frac{1}{2}$ is a 1st order pole of $\cot((m + \frac{1}{2})z/\tau)$ and thus is a 1st order pole of $g_m(z)$

3. $\tan(0) = 0$. Thus 0 is a pole of g_m . Let $n = m + \frac{1}{2}$

$$\begin{aligned}g_m(z) &= \frac{\cos(nz) \cos(nz/\tau)}{z \left(\sum_{j=0}^{\infty} \frac{(-n)^j z^{2j+1}}{(2j+1)!} \right) \left(\sum_{j=0}^{\infty} \frac{(-n/\tau)^j z^{2j+1}}{(2j+1)!} \right)} \\ &= \frac{\cos(nz) \cos(nz/\tau)}{z^3 \left(\sum_{j=0}^{\infty} \frac{(-n)^j z^{2j}}{(2j+1)!} \right) \left(\sum_{j=0}^{\infty} \frac{(-n/\tau)^j z^{2j}}{(2j+1)!} \right)}\end{aligned}$$

$\left(\sum_{j=0}^{\infty} \frac{(-n)^j z^{2j}}{(2j+1)!} \right) \left(\sum_{j=0}^{\infty} \frac{(-n/\tau)^j z^{2j}}{(2j+1)!} \right) = \frac{n^2}{\tau} \neq 0$. Thus

$$\frac{\cos(nz) \cos(nz/\tau)}{\left(\sum_{j=0}^{\infty} \frac{(-n)^j z^{2j}}{(2j+1)!} \right) \left(\sum_{j=0}^{\infty} \frac{(-n/\tau)^j z^{2j}}{(2j+1)!} \right)}$$

is holomorphic in a neighborhood around 0.

Thus 0 is a order 3 pole of g_m .

□

Problem 2d.

Proof. Let $n = m + \frac{1}{2}$

1. For $k \in \mathbb{Z} \setminus \{0\}$

$$\begin{aligned} \text{Res}_{\frac{k\pi}{n}}(g_m) &= \frac{n \cos(k\pi) \cot(k\pi/\tau)}{k\pi \left(\sum_{j=0}^{\infty} \frac{(-n)^{j+k} (x - \frac{k\pi}{n})^{2n}}{(2n+1)!} \right) \Big|_{x=\frac{k\pi}{n}}} \\ &= \frac{n(-1)^k \cot(k\pi/\tau)}{k\pi(-n)^k} \\ &= \frac{\cot(k\pi/\tau)}{k\pi} \end{aligned}$$

2. For $k \in \mathbb{Z} \setminus \{0\}$

$$\begin{aligned} \text{Res}_{\frac{k\pi\tau}{n}}(g_m) &= \frac{n \cos(k\pi) \cot(k\pi\tau)}{k\tau\pi \left(\sum_{j=0}^{\infty} \frac{(-n/\tau)^{j+k} (x - \frac{k\pi\tau}{n})^{2n}}{(2n+1)!} \right) \Big|_{x=\frac{k\pi\tau}{n}}} \\ &= \frac{n(-1)^k \cot(k\pi\tau)}{k\pi\tau(-n/\tau)^k} \\ &= \frac{\cot(k\pi\tau)}{k\pi} \end{aligned}$$

3. Note $\cot(z) = \frac{1}{z} + \frac{z}{3} + O(z^3)$.

So

$$\begin{aligned} g_m(z) &= \frac{1}{z} \cot(nz) \cot(nz/\tau) = \frac{1}{z} \left(\frac{1}{nz} + \frac{nz}{3} + O(z^3) \right) \left(\frac{\tau}{nz} + \frac{nz}{3\tau} + O(z^3) \right) \\ &= \frac{1}{z^3} \left(\frac{1}{n} + \frac{nz^2}{3} + O(z^4) \right) \left(\frac{\tau}{n} + \frac{nz^2}{3\tau} + O(z^4) \right) \\ &= \frac{1}{z^3} \left(\frac{\tau}{n^2} + \frac{z^2}{3\tau} + \frac{z^2\tau}{3} + O(z^3) \right) \end{aligned}$$

Thus

$$\begin{aligned}
Res_0(g_m) &= \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^2 g_m(z) \\
&= \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(\frac{\tau}{n^2} + \frac{z^2}{3\tau} + \frac{z^2\tau}{3} + O(z^3) \right) \\
&= \lim_{z \rightarrow 0} \left(\frac{1}{3\tau} + \frac{\tau}{3} + O(z) \right) \\
&= \frac{1}{3}(\tau + \tau^{-1})
\end{aligned}$$

□

Problem 2e.

Proof. WLOG $\tau \in \mathcal{H}$. Let $n = \frac{1}{2} + m$.
Since

$$\cot(nz) = i \left(\frac{2}{1 - e^{-2inz}} - 1 \right) = i \left(\frac{2}{1 - e^{-2in\Re(z)} e^{2in\Im(z)}} - 1 \right)$$

as proven above we see that, as $\Im(nz) \rightarrow -\infty$, $e^{2in\Im(z)} \rightarrow 0$ so $\cot(nz) = i$. as $\Im(z) \rightarrow \infty$, $e^{2in\Im(z)} \rightarrow \infty$ so $\cot(nz) \rightarrow -i$.

Thus for large m , hence large n , if $\Im(nz) \gg 0 \implies \cot nz \approx -i$, otherwise if $\Im(nz) \ll 0 \implies \cot nz \approx i$. Similar analysis applies for $\cot nz/\tau$, where you look at the sign $\Im(nz/\tau)$ instead.

$\forall a, b \in \mathbb{C}$, let $[a, b]$ be the line segment from a to b and let $(a, b) = [a, b] \setminus \{a, b\}$.

$\forall z \in [1, \tau]$, the sign of $\Im(nz)$ equals the sign of $\Im(n\tau)$, thus the value of $\cot(nz) \approx -i$ or $\cot nz \approx i$ throughout the edge. Likewise, the sign of $\Im(nz/\tau)$ is the same as sign of $\Im(\frac{n}{\tau})$. Thus the value of $\cot(nz) \approx -i$ or $\cot nz \approx i$ throughout the edge.

Thus

$$f_m(z) \rightarrow \pm 1 \text{ for } z \in [1, \tau]$$

meaning

$$g_m(z) \rightarrow \pm \frac{1}{z} \text{ for } z \in [1, \tau]$$

Similar analysis applies for $[\tau, -1], [-1, -\tau], [-\tau, 1]$.

Not, $1, \tau, -1, -\tau$ are not any poles, so $f_m(z)$ and $g_m(z)$ don't blow up.

Note since no pole is included in Γ , each g_m is bounded. $|g_m(z)|$ stays close to $|\frac{1}{z}|$. At the vertices no poles of g_m occur.

Therefore g_m is uniformly bounded on Γ .

Thus $\cot(nz) \approx -i$ for $z \in [1, \tau] \cup [\tau, -1]$, and $\cot(nz/\tau) = \cot(nz\bar{\tau}/|\tau|^2) \approx i$ if $\Im(z\bar{\tau}) < 0$.

Note

$$\Im(z\bar{\tau}) = \Im(z)\Re\tau - \Re z\Im(\tau) < 0 \iff \Im(z) < \Re z \frac{\Im(\tau)}{\Re\tau}$$

Thus $\cot(nz/\tau) \approx i$ for $z \in [1, \tau] \cup [-\tau, 1]$.

Thus $f_m(z) = 1$ for $z \in [1, \tau] \cup [-1, \tau]$ and $f_m(z) = -1$ for $z \in [\tau, -1] \cup [-\tau, 1]$.

Thus by dominated convergence theorem

$$\begin{aligned}
\lim_{m \rightarrow \infty} \int_{\Gamma} g_m(z) dz &= \int_{\Gamma} \lim_{m \rightarrow \infty} g_m(z) dz \\
&= \int_{\Gamma} \lim_{m \rightarrow \infty} f_m(z) \frac{dz}{z} \\
&= \left(\int_1^{\tau} - \int_{\tau}^{-1} + \int_{-1}^{-\tau} - \int_{-\tau}^1 \right) \frac{1}{z} dz \\
&= 2 \left(\int_1^{\tau} - \int_{-\tau}^1 \right) \frac{1}{z} dz \\
&= 2(\log(\tau) - \log(1) - \log(-\tau) + \log(1)) \\
&= 2(\log(\tau) - \log(-\tau)) \\
&= 4(\log(\tau) - \pi \frac{i}{2}) = 4 \log(\tau/i)
\end{aligned}$$

Thus

$$\begin{aligned}
\exp(3 \lim_{m \rightarrow \infty} \int_{\Gamma} g_m(z) dz) &= \exp(12 \log(\tau/i)) \\
&= \exp(\log(\tau/i))^{12} \\
&= \tau^{12}
\end{aligned}$$

□

Problem 2f.

Proof. By the residue theorem,

$$\begin{aligned}
\int_{\Gamma} g_m(z) dz &= \frac{-2\pi i (\tau + \tau^{-1})}{3} + 2\pi i \left(\sum_{i=1}^m \frac{2}{\pi} (\cot(k\pi\tau) + \cot(k\pi/\tau)) \right) \\
&= \frac{-2\pi i (\tau + \tau^{-1})}{3} + 4i \left(\sum_{i=1}^m -i \left(1 - \frac{2}{1 - e(k\tau)} + 1 - \frac{2}{1 - e(k/\tau)} \right) \right) \\
&= \frac{-2\pi i (\tau + \tau^{-1})}{3} - 8 \left(\sum_{i=1}^m \left(\frac{1}{1 - e(k\tau)} + \frac{1}{1 - e(k/\tau)} \right) \right)
\end{aligned}$$

□

Problem 2g.

Proof.

$$\begin{aligned}
\log \left(\frac{\Delta(-\frac{1}{\tau})}{\Delta(\tau)} \right) &= \log \left(\Delta(-\frac{1}{\tau}) \right) - \log(\tau) \\
&= \left(\log \left(e(-\frac{1}{\tau}) \right) + 24 \sum_{k=1}^{\infty} \log(1 - e(-k/\tau)) \right) - \log(e(\tau)) + 24 \sum_{k=1}^{\infty} \log(1 - e(-k/\tau)) \\
&= 3 \int_{\Gamma} g_m(z) dz
\end{aligned}$$

Since the LHS doesn't depend on m ,

$$\frac{\Delta(-\frac{1}{\tau})}{\Delta(\tau)} = \exp(3 \lim_{m \rightarrow \infty} \int_{\Gamma} g_m(z) dz) = \tau^{12}$$

Note

$$\Delta\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \tau\right) = \Delta(-\frac{1}{\tau}) = \tau^{12} \Delta(\tau)$$

Moreover since $\Delta \in M_{12}$,

$$\Delta\left(\begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix} \tau\right) = \Delta(\tau + 1) = (0\tau + 1)^{12} \Delta(\tau) = \Delta(\tau)$$

Since both matrices generate $PSL_2(\mathbb{Z})$ it follows that $\Delta \in M_{12}^o$. □